

Managing Time Series Data Using InfluxData

InfluxData is used when IoT deployments require the support of data from a large number, say thousands, of sensors. This data can be collected, stored and used to issue alerts. It is a good choice when it comes to building custom monitoring solutions.



In a typical IoT application, data is collected at servers and middleware solutions from many devices with multiple sensors, and each device cluster generates a high volume of data at the rate of thousands to millions of records per second. These records are mapped to a time scale with nano second precision. Even though traditional databases are highly efficient for typical rates of insertions and deletions, they may not be able to handle high precision data optimally. Moreover, it is very rare to delete individual records in IoT or DevOps scenarios, and updates are not allowed most of the time. Instead, data may be dropped for a particular duration periodically after taking aggregates or anomalies. So, specialised storage engines optimised for this purpose are required.

A time series data base is an optimised solution for the above challenges. The list below gives open source databases designed for time series data management; some

are exclusively designed for time series needs, and a few* come with additional support like custom storage engines and schema considerations:

- InfluxDb, a part of InfluxData TICK stack, with independent storage engine
- OpenTSDB, on top of Apache HBase/Cassandra or Google's BigTable
- KairosDB, on top of Apache Cassandra
- Riak TS from Basho – NoSQL Time Series
- Whisper Database, part of Graphite tool
- Round Robin Database (RRD) tool
- Prometheus.io - designed for monitoring metrics
- MongoDB*
- PostgreSQL*

In this article, let's take a tour of the InfluxData TICK stack and two of its components —Telegraf for data collection and InfluxDb for data storage. InfluxDb has